



Final Report Project
(Counselling Session Booking System – Group 16)
CSE3023 Web-Based Application Development
Bachelor of Computer Science (Software Engineering)

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Github link: <https://github.com/ForStudy111/WebAssignmentGroup16.git>

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I. Group Members, Leader and Module Task Distribution

The project modules are distributed according to the system workflow. Each member is responsible for one specific module with CRUD functionality.

Member Name	Assigned Module	Module Functions (CRUD)
WAI PEI YUEN (Group Leader)	User Management, Schedule Management	User Management Create: register account View: login account Update: update profile Delete: delete account Schedule Management Create: create schedule View: view schedule Update: update schedule Delete: delete schedule
TENGKU AQIL BIN TENGKU AMDAN	Counsellor Management	Counsellor Management Create: add counsellor View: view counsellor list Update: update counsellor list Delete: remove counsellor
AMERRUL RAFIQUE BIN MOHAMED SULAIMAN	Booking Management	Booking Management Create: book session View: view bookings Update: update booking (reschedule) Delete: cancel booking

MUHAMMAD ADAM DANISH BIN MOHAMAD FAIZ	Session Record Management	Session Record Management Create: add session notes and feedback View: view session history Update: edit notes Delete: remove records
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Table I: Group Number: Group 16 Group Members, Leader and Module Task Distribution

Table I.I shows the distribution of modules in our group. All modules have it own CRUD operations.

1.0 Introduction

1.1 Project background

Counseling services play a very important role in helping students deal with various academic, emotional, social and personal challenges throughout their studies. However, the counseling session management process that is still done manually often causes various problems such as overlapping schedules, difficulty in managing records, communication delays and inefficiency in booking management.

Along with the development of information and communication technology, the use of web-based systems has become a necessity in improving the efficiency of management of an organization. Therefore, the Counseling Session Booking System project was developed to provide a centralized platform that allows students to book counseling sessions online more easily, quickly and systematically.[1][3]

This system was developed using Java Web technology based on Model-View-Controller (MVC) architecture which separates application logic, user interface and data management to ensure the system is more organized and easy to maintain. This system involves three main categories of users which are students, counselors and administrators. Each user is given access to certain functions according to their respective roles.[4]

Through this system, students can view available counselor schedules, book sessions, manage bookings that have been made and provide feedback after the session is complete. Counselors can manage their availability schedule, review received bookings, approve sessions and record counseling reports. Administrators are responsible for managing user and counselor accounts to ensure the system operates smoothly.

In conclusion, the counseling session booking system is the process that can be done more efficiently, systematically and professionally compared to the manual methods used previously.

1.2 Problem statement

Currently, the process of booking counseling sessions is often done manually via phone calls, emails or messaging apps. This method can lead to duplicate bookings, scheduling conflicts, lost records and communication delays between students and counselors.[2]

In addition, manual record management makes it difficult for counselors and administrators to track session history and effectively monitor counseling activities. Therefore, a centralized system is needed to manage counseling session bookings more systematically, reduce errors and improve management efficiency.

1.3 Project objectives

The objectives of this project are:

1. To develop a web-based counseling session booking system that is easy to use by students, counselors and administrators.
2. To provide a platform that allows students to view available counselor schedules and book sessions online.
3. To allow counselors to manage availability schedules, review bookings and systematically record counseling session information.
4. To allow administrators to manage counselor information and monitor bookings.
5. To allow counselors to record session notes and students to submit feedback sessions.
6. To provide centralized storage of session records and user feedback for reference and monitoring purposes.
7. To reduce the use of manual methods in counseling session management to increase productivity and data accuracy.
8. To reduce invalid input and schedule conflicts through validation.

1.4 Scope of project

The scope of the Counseling Session Booking System outlines the specific boundaries, target users and functional and non-functional requirements implemented within the web application.

1.4.1 Target Users and Roles

The system is bounded to serve three distinct groups of authenticated users within the institution, each provisioned with role-based access control:

- **Students:** Users who can navigate available counseling slots, book new appointments, manage their pending/active sessions and provide feedback after a completed evaluation.
- **Counselors:** Users responsible for creating and updating their availability schedules, reviewing and approving student appointment bookings and updating official post-session case records.
- **Administrators:** Superusers tasked with system governance included the provisioning and structural management (CRUD operations) of students(only can be Read and Update) and counselor accounts(Full CRUD).

1.4.2 Function Scope

The functional boundaries of the application focus entirely on automating the scheduling lifecycle via a clean web interface. This covers:

- **Authentication & Profile Management:** Secure login capabilities validating credentials against a centralized database schema, paired with self-service profile editing and password resets.
- **Interactive Scheduling Matrix:** Self-service time slots booking where counselor-defined time slots are dynamically rendered for student selection to mitigate double-booking conflicts.
- **Appointment Lifecycle Tracking:** State-driven session handling allowing real-time scheduling updates, student-initiated cancellations and formal counselor approval checkpoints.

- Feedback & Documentation Loop: Secure archival pipelines for students to submit post-session feedback reviews and for counselors to log diagnostic case reports into permanent storage.

1.4.3 Technical and Architectural Scope

- The system is engineered following the robust Model-View-Controller (MVC) software architectural pattern to separate underlying business rules from presentation elements.
- The implementation is strictly limited to an environment comprising Java Web Servlets for handling controller logic, JavaServer Pages (JSP) for dynamic view rendering and MySQL (managed via phpMyAdmin) over a JDBC pipeline for structural data persistence.
- Deployment hosting boundaries are configured using a local standalone instances framework via XAMPP and the Apache Tomcat web engine.

1.4.4 Out-of-Scope (System Limitations)

To keep project development measurable and efficient, the following functionalities are explicitly designated as out-of-scope:

- Automated third-party communications or multi-channel alert sequences (such as SMS or email notification alerts).
- Real-time video conferencing engine or inline virtual meeting room infrastructure (such as direct native API integrations with Zoom or Google Meet platforms).
- Advanced business intelligence reporting structures or data analysis metrics graphs for administrators.

2.0 System Design

2.1 Use Case Diagram

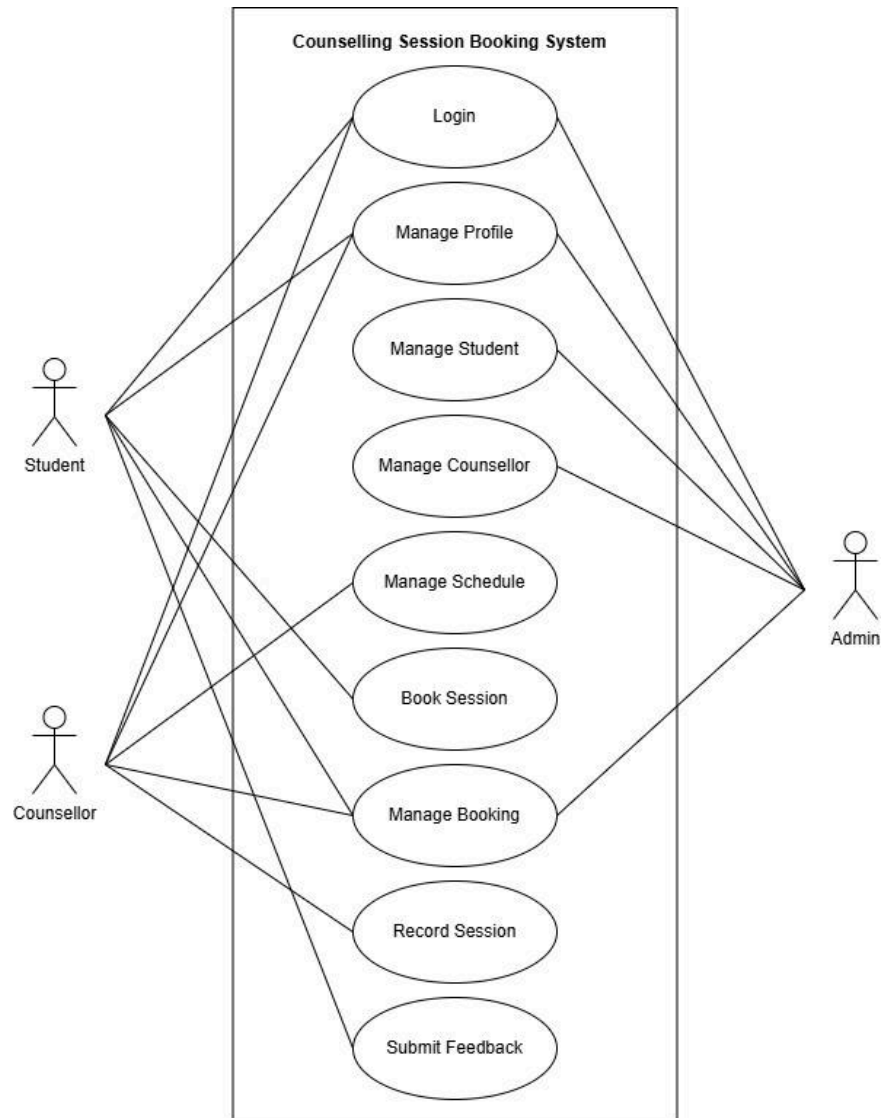


Figure 2.1: Use Case Diagram of Counseling Session Booking System

Figure 2.1 shows the overall use case diagram of the Counseling Session Booking System. The diagram illustrates the interaction between the three main actors (Student, Counsellor and Administrator) with the system. Each actor is assigned specific functionalities according to their role and responsibilities within the system.

2.2 Use Case Description

2.2.1 Login

Use Case Name: Login	ID: 1	Importance Level: High
Primary Actor: Student, Counsellor, Admin	Use Case Type: Essential	
Stakeholders and Interests: Student – Wants to access the system features Counsellor – Wants to manage schedule and sessions Admin – Wants to manage system operations		
Brief Description: This use case describes how users log in to the system using valid credentials.		
Trigger: User selects Login on the system homepage Type: External		
Relationships: Association: Student, Counsellor, Admin Include: None Extend: None Generalization: None		
Normal Flow of Events: 1. The User enters username and password. 2. The System validates the credentials. 3. The System grants access to the dashboard.		
SubFlows:		
Alternate/Exceptional Flows: 2a: Invalid credentials The system displays an error message and prompts re-entry.		

Table 2.2.1: Login Use Case Description

Table 2.2.1 shows the foundational login logic. The system validates the username and password based on the database to ensure only authorized users access the dashboard. If not authorized, an "Invalid username or password" message will be sent.

2.2.2 Manage Profile

Use Case Name: Manage Profile	ID: 2	Importance Level: Medium
Primary Actor: Student, Counsellor, Admin	Use Case Type: Essential	
Stakeholders and Interests: All actors – Wants to update their personal information easily		
Brief Description: This use case describes how users view and update their profile details.		
Trigger: User selects “Manage Profile” Type: External		
Relationships: Association: Student, Counsellor, Admin Include: None Extend: None Generalization: None		
Normal Flow of Events: 1. The Actor selects the profile section. 2. The System displays current information. 3. The Actor updates details. 4. The Actor submits changes. 5. The System validates and saves data.		
SubFlows:		
Alternate/Exceptional Flows: 5a: Invalid input The system displays an error message.		

Table 2.2.2: Manage Profile Use Case Description

Table 2.2.2 shows the manage profile logic. It allows all actors to keep and update their personal information with validation logic to prevent the submission of invalid input data.

2.2.3 Manage Student

Use Case Name: Manage Student	ID: 3	Importance Level: High
Primary Actor: Admin	Use Case Type: Essential	
Stakeholders and Interests: Admin – Wants to view student information and update student’s details.		
Brief Description: This use case describes how the Admin views student information and updates student’s details.		
Trigger: Admin selects “Student List” Type: External		
Relationships: Association: Admin Include: None Extend: None Generalization: None		
Normal Flow of Events: 1. The Admin opens the Student List Page. 2. The System displays all student records. 3. The Admin selects a student record. 4. The Admin updates the student’s details (e.g. phone number). 5. The system validates and saves the student details.		
SubFlows:		
Alternate/Exceptional Flows: 5a: Invalid phone number		

The system displays an error message and does not save the changes.

Table 2.2.3: Manage Student Use Case Description

Table 2.2.3 shows the manage student logic. It focuses on maintaining the accuracy of student contact information such as phone numbers, the other student will be readonly to keep data accurate.

2.2.4 Manage Counsellor

Use Case Name: Manage Counsellor	ID: 4	Importance Level: High
Primary Actor: Admin	Use Case Type: Essential	
Stakeholders and Interests: Admin – Wants to manage counsellor information		
Brief Description: This use case describes how the Admin manages counsellor accounts.		
Trigger: Admin selects “Manage Counsellor” Type: External		
Relationships: Association: Admin Include: None Extend: None Generalization: None		
Normal Flow of Events: 1. The Admin views counsellor list. 2. The Admin adds, updates or deletes counsellor data. 3. The system saves the changes.		
SubFlows:		
Alternate/Exceptional Flows:		

Table 2.2.4: Manage Counsellor Use Case Description

Table 2.2.4 shows a Manage Counsellor logic. This use case provides CRUD (Create, Read, Update, Delete) operation capabilities to the Admin for managing the counselling staff.

2.2.5 Manage Schedule

Use Case Name: Manage Schedule	ID: 5	Importance Level: High
Primary Actor: Counsellor	Use Case Type: Essential	
Stakeholders and Interests: Counsellor – Wants to manage availability schedule		
Brief Description: This use case describes how the Counsellor sets and updates available time slots.		
Trigger: Counsellor selects “Manage Schedule” Type: External		
Relationships: Association: Counsellor Include: None Extend: None Generalization: None		
Normal Flow of Events: 1. The Counsellor views the current availability slots. 2. The Counsellor creates updates or deletes an unused slot. 3. The system checks for schedule overlap. 4. The system saves the schedule changes.		
SubFlows:		
Alternate/Exceptional Flows: 3a: Overlapping the time slot The system display an error message and does not save the slot.		

Table 2.2.5: Manage Schedule Use Case Description

Table 2.2.5 shows the Manage Schedule logic. This use case is essential for operational efficiency. It includes a business rule that checks for overlapping time slots, preventing the counsellor from being double-booked and ensuring only valid slots are saved.

2.2.6 Book Session

Use Case Name: Book Session	ID: 6	Importance Level: High
Primary Actor: Student	Use Case Type: Essential	
Stakeholders and Interests: Student – Wants to book a counselling session Counsellor – Wants organized schedule		
Brief Description: This use case describes how the Student books a counselling session.		
Trigger: Student selects available schedule Type: External		
Relationships: Association: Student Include: None Extend: None Generalization: None		
Normal Flow of Events: 1. The Student views the available counseling schedule. 2. The Student selects a time slot. 3. The Student confirms the booking request. 4. The System creates the booking with PENDING status.		
SubFlows:		
Alternate/Exceptional Flows: 2a: Slot unavailable The system displays messages.		

Table 2.2.6: Book Session Use Case Description

Table 2.2.6 shows the Book Session logic. This is the core transactional use case. The student views available times and confirms a selection. The system sets the booking status to "PENDING" initially, then waits for the counsellor for final approval.

2.2.7 Manage Booking

Use Case Name: Manage Booking	ID: 7	Importance Level: Medium
Primary Actor: Student, Counsellor, Admin	Use Case Type: Essential	
Stakeholders and Interests:		
Student – Wants to reschedule or cancel a booking		
Counsellor – Wants to manage session appointments		
Admin – Wants to oversee bookings		
Brief Description:		
This use case describes how bookings are managed.		
Trigger: User selects “Manage Booking”		
Type: External		
Relationships:		
Association: Student, Counsellor, Admin		
Include: None		
Extend: None		
Generalization: None		
Normal Flow of Events:		
1. The Actor views the relevant booking list.		
2. A Student may reschedule or cancel a booking.		
3. A Counsellor may approve, reject or cancel a booking.		
4. An Admin may view and monitor booking records.		
5. The system saves the updated booking status.		
3. The System save changes.		
SubFlows:		

Alternate/Exceptional Flows:

Table 2.2.7: Manage Booking Use Case Description

Table 2.2.7 shows the Manage Booking logic. This use case handles the lifecycle of an appointment. It allows for multi-actor interaction in which students can manage their own requests (reschedule/cancel) while counsellors have the authority to approve or reject requests.

2.2.8 Record Session

Use Case Name: Record Session	ID: 8	Importance Level: Medium
Primary Actor: Counsellor	Use Case Type: Essential	
Stakeholders and Interests: Counsellor – Wants to record session details		
Brief Description: This use case describes how the Counsellor records session notes.		
Trigger: Session completed Type: External		
Relationships: Association: Counsellor Include: None Extend: None Generalization: None		
Normal Flow of Events: 1. The Counsellor selects a completed session. 2. The Counsellor enters session notes. 3. The System saves the session record.		
SubFlows:		
Alternate/Exceptional Flows:		

Table 2.2.8: Record Session Use Case Description

Table 2.2.8 shows the Record Session logic. This logic makes sure the proper documentation for each counseling session. Once a session is completed, the counsellor adds private session notes to the record.

2.2.9 Submit Feedback

Use Case Name: Submit Feedback	ID: 9	Importance Level: Medium
Primary Actor: Student	Use Case Type: Essential	
Stakeholders and Interests: Student – Wants to give feedback Admin – Wants to improve service		
Brief Description: This use case describes how the Student submits feedback.		
Trigger: Student selects completed session Type: External		
Relationships: Association: Student Include: None Extend: None Generalization: None		
Normal Flow of Events: 1. The Student selects a session. 2. The Student enters ratings and comments. 3. The System saves feedback.		
SubFlows:		
Alternate/Exceptional Flows:		

Table 2.2.9: Submit Feedback Use Case Description

Table 2.2.9 shows the Submit Feedback logic. Once a session is completed and the counselor submitted a session record, students can submit ratings and comments like a suggestion to improve further counseling service or what they want to say to the counselor.

2.3 Class Diagram

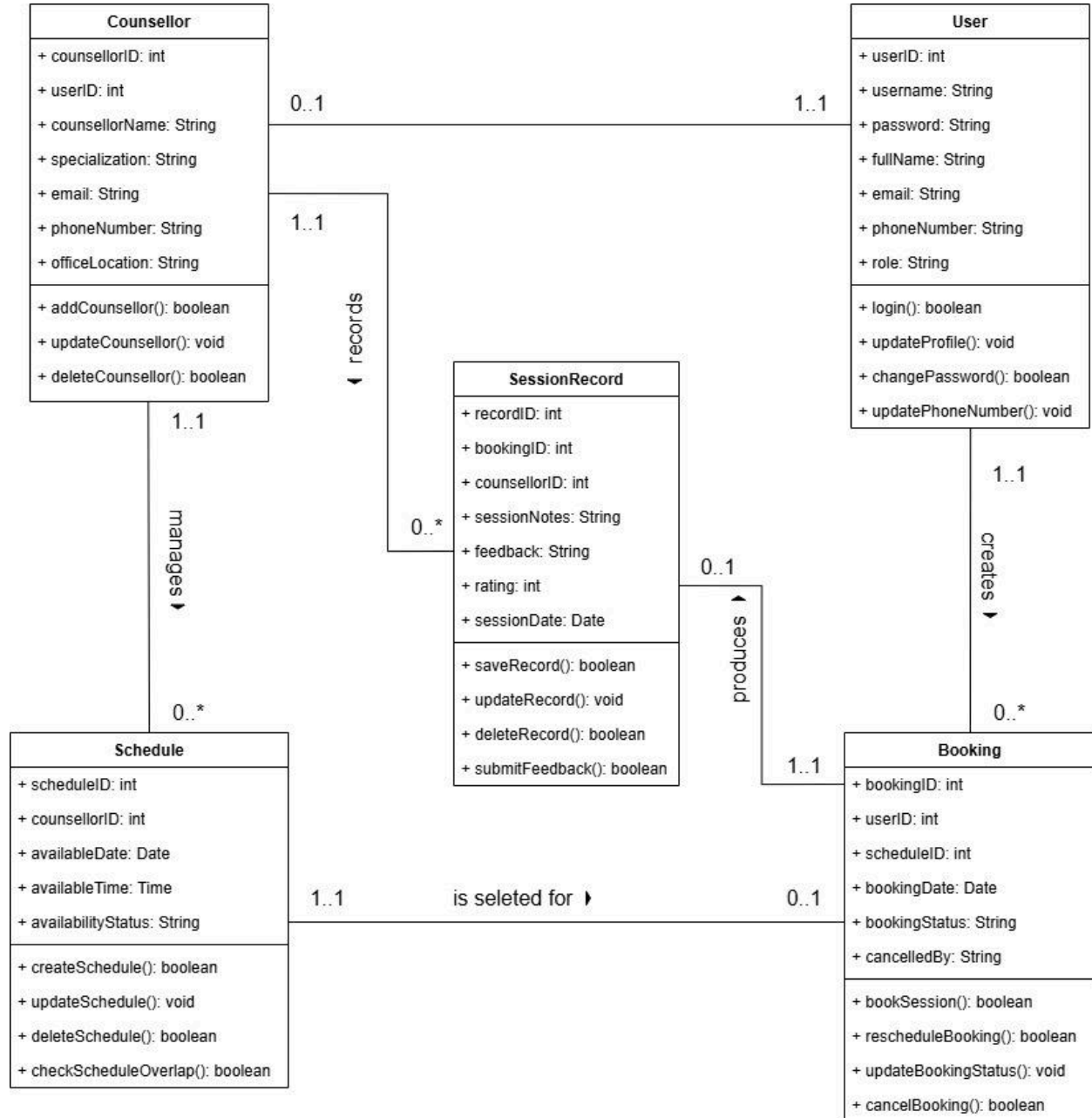


Figure 2.3: Class Diagram of Counseling Session Booking System

Figure 2.3 shows the class diagram of the Counseling Session Booking System. The diagram illustrates the relationships between system entities such as User, Counsellor, Schedule, Booking and Session Record. It also demonstrates how the classes interact to support the system functionalities.

2.4 Database Planning (Model)

The database planning identifies all required data to be stored in the system together with the specific data types.

Table 2.4.1: users

Attribute	Data Type	Key	Description
user_id	INT	PK	Unique user ID
username	VARCHAR(50)	-	Username for login
password	VARCHAR(255)	-	User password
full_name	VARCHAR(100)	-	User full name
email	VARCHAR(100)	-	User email
phone_number	VARCHAR(20)	-	Contact number
role	VARCHAR(20)	-	User role (Student/Counselor/Admin)

Table 2.4.1: User Data Database

Table 2.4.1 is the table that stores the foundational information for all system users. The user_id acts as the Primary Key (PK) to ensure every user is uniquely identifiable and while the role attribute differentiates the access levels between students, counselors and administrators.

Table 2.4.2: counsellors

Attribute	Data Type	Key	Description
counsellor_id	INT	PK	Unique counsellor ID
user_id	INT	FK	References users
counsellor_name	VARCHAR(100)	-	Counsellor name
specialization	VARCHAR(100)	-	Counselling specialization
email	VARCHAR(100)	-	Counsellor email
phone_number	VARCHAR(20)	-	Contact number
office_location	VARCHAR(100)	-	Office location

Table 2.4.2: Counsellors Data Database

Table 2.4.2 is the table that maintains counsellorspecific data. It uses user_id as a Foreign Key (FK) to link the professional profile to the base user account. This separation can make sure that sensitive scheduling and specialization data remains organized under the correct role.

Table 2.4.3: schedules

Attribute	Data Type	Key	Description
schedule_id	INT	PK	Unique schedule ID
counsellor_id	INT	FK	References counsellor
available_date	DATE	-	Available session date
available_time	TIME	-	Available session time
availability_status	VARCHAR(20)	-	Slots status such as AVAILABLE or BOOKED

Table 2.4.3: Schedules Data Database

Table 2.4.3 is the table that stores the time slots created by counsellors. By linking to the counsellor_id with a FK, the system can track which staff member is available at a specific date and time. The availability_status is critical for preventing double-booking by updating the slot state once a booking is confirmed.

Table 2.4.4: bookings

Attribute	Data Type	Key	Description
booking_id	INT	PK	Unique booking ID
user_id	INT	FK	References user
schedule_id	INT	FK	References schedule
booking_date	DATE	-	Date of booking
booking_status	VARCHAR(20)	-	Booking status
cancelled_by	VARCHAR(20)	-	Role that cancelled the booking

Table 2.4.4: Bookings Data Database

Table 2.4.4 is the table that centralizes appointment information. It captures the relationship between a specific user and a schedule slot. The cancelled_by attribute provides auditing capability and documents whether a cancellation was initiated by a student, counsellor or admin.

Table 2.4.5: session_records

Attribute	Data Type	Key	Description
record_id	INT	PK	Unique session record ID
booking_id	INT	FK	References booking
counsellor_id	INT	FK	References counsellor
session_notes	TEXT	-	Counselling session notes
feedback	TEXT	-	User feedback
rating	INT	-	User rating score
session_date	DATE	-	Date of counselling session

Table 2.4.5: Session Record Data Database

Table 2.4.5 is the table used for post-session documentation. It links to both the booking_id and counsellor_id to ensure that session notes and student feedback are correctly associated with the appropriate appointment and counselor. This provides the system with a complete history for student wellbeing tracking.

2.5 Sitemap / Navigation Structure

The following is a sitemap for the hierarchical structure, pages and relationships of the website. It acts as an organizational directory that helps design the user navigation.(figure 2.5)

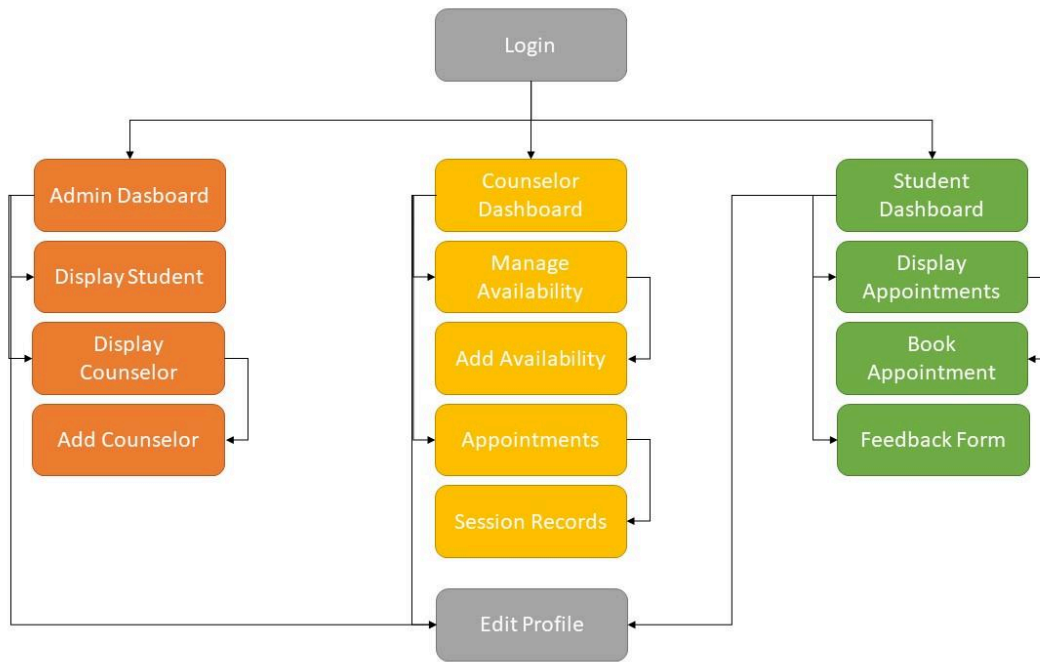


Figure 2.5: Sitemap

Figure 2.5 shows the sitemap of the Counseling Session Booking System. The sitemap illustrates the navigation hierarchy and relationships between pages available to administrators, counsellors and students.

2.6 Wireframe or User Interface Design

The following figures (2.6.1 until 2.6.12) are the initial outline of the web interface. It depicts the layout, structure and functional placement before any detailed graphic design is added. The functional descriptions of the pages can be referred to in section 2.7.

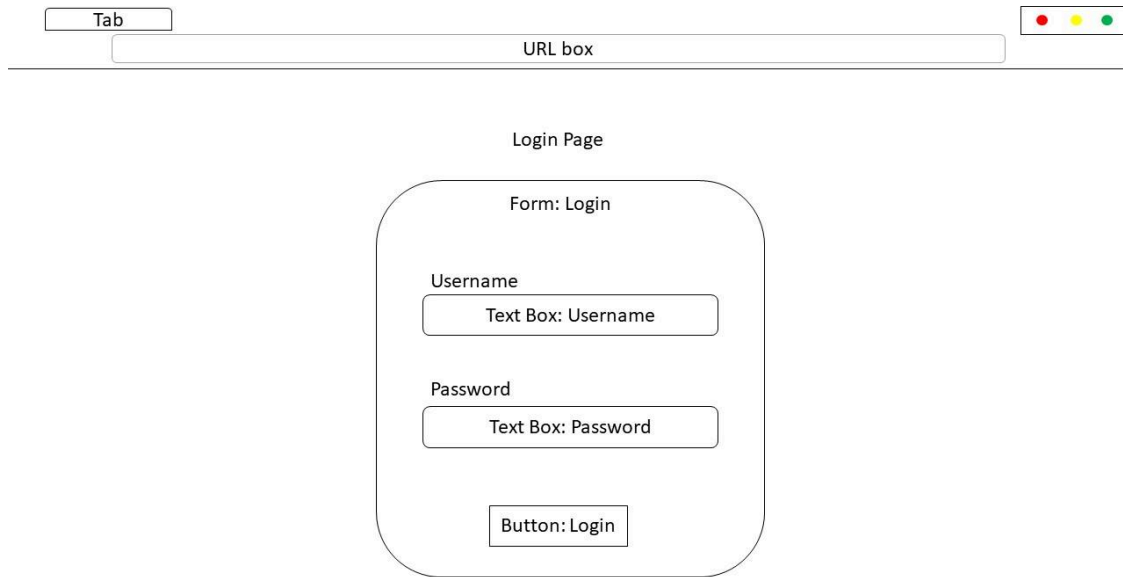


Figure 2.6.1: Login Page

Figure 2.6.1 shows the Login Page wireframe. This page allows users to enter their credentials and access the system according to their assigned roles.

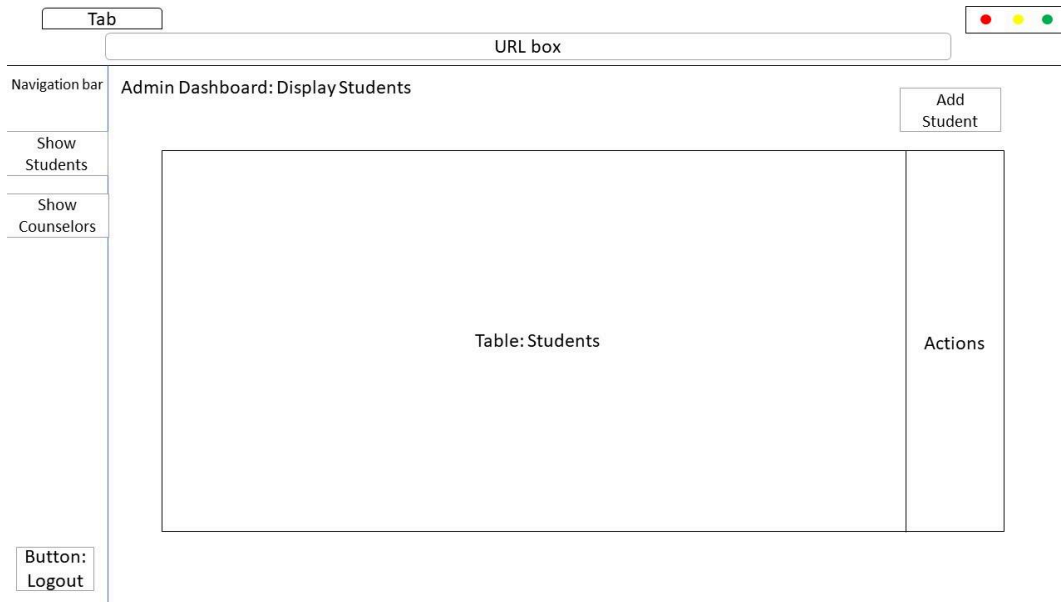


Figure 2.6.2: Student Page

Figure 2.6.2 shows the Student Management Page. The page enables administrators to view student information and update selected profile details.

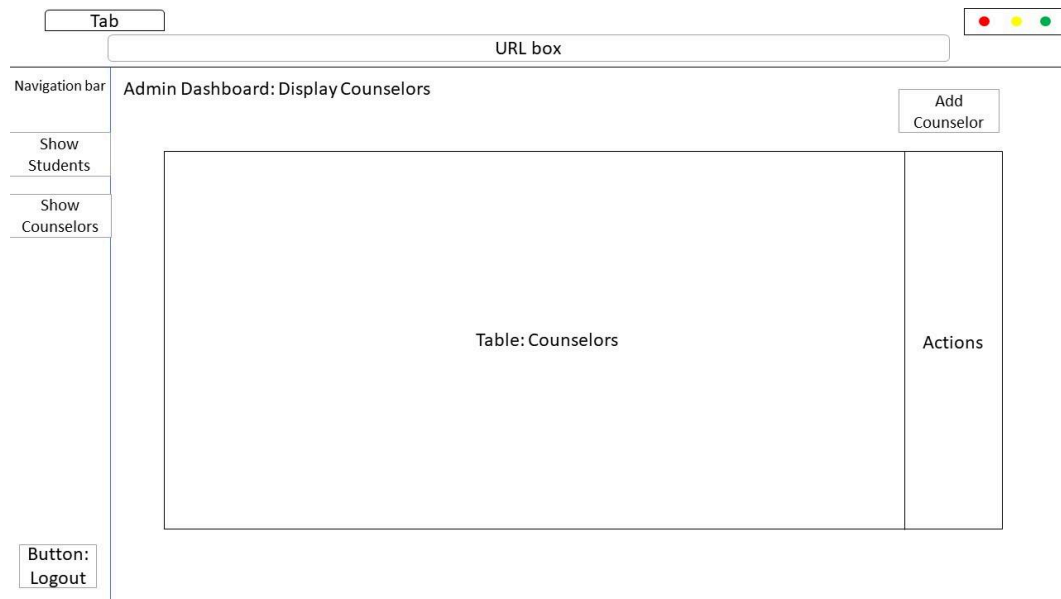
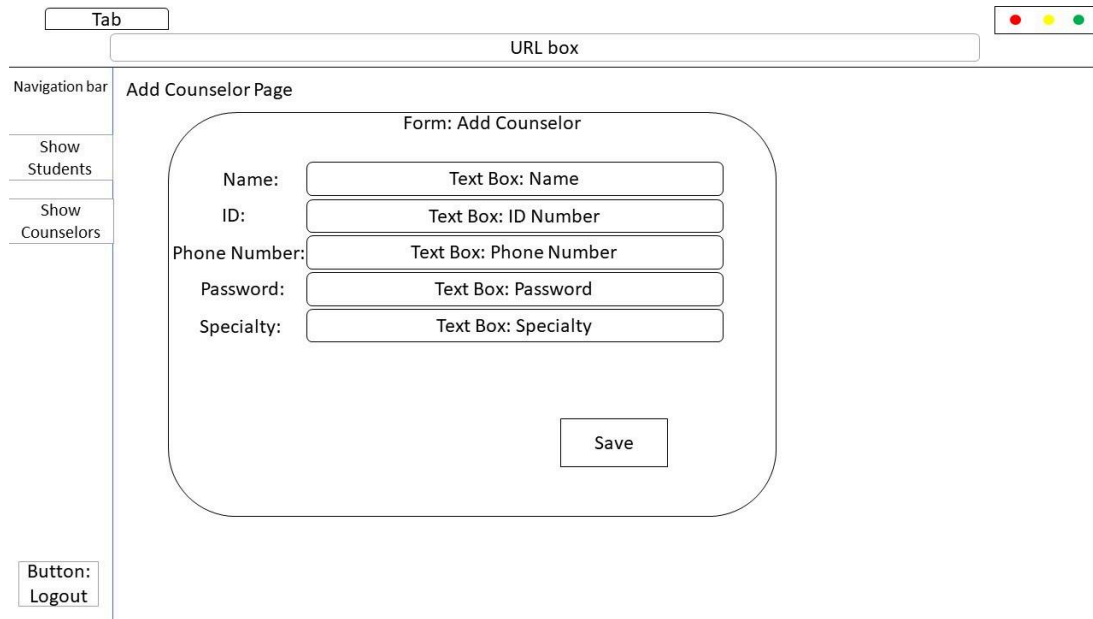


Figure 2.6.3: Display Counselor Page

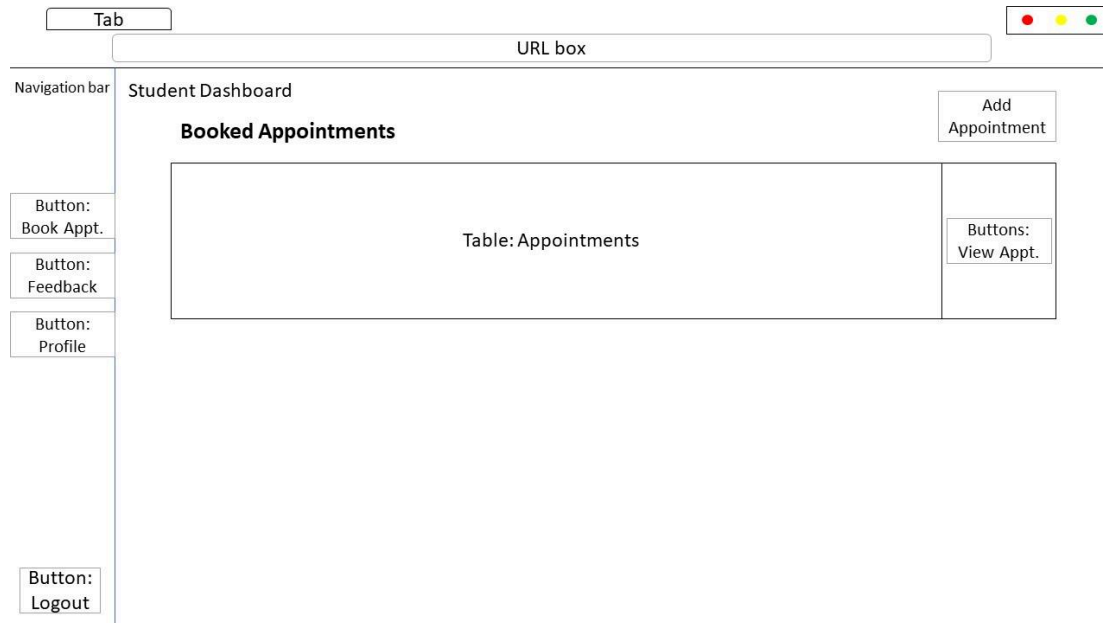
Figure 2.6.3 shows the Counsellor Management Page. Administrators can view counsellor records and perform account management functions.



The image shows a web browser window with a single tab. The address bar contains a 'URL box'. The page title is 'Add Counselor Page'. On the left, there is a 'Navigation bar' with two links: 'Show Students' and 'Show Counselors'. Below the navigation bar is a 'Button: Logout'. The main content area is titled 'Form: Add Counselor' and contains five text input fields with labels: 'Name:', 'ID:', 'Phone Number:', 'Password:', and 'Specialty:'. Each field is followed by a text box labeled 'Text Box: [Field Name]'. A 'Save' button is located at the bottom right of the form.

Figure 2.6.4: Add Counselor Page

Figure 2.6.4 shows the Add Counsellor interface. This page allows administrators to register new counsellors into the system.



The image shows a web browser window with a single tab. The address bar contains a 'URL box'. The page title is 'Student Dashboard'. On the left, there is a 'Navigation bar' with four links: 'Book Appt.', 'Feedback', 'Profile', and 'Logout'. The main content area is titled 'Booked Appointments'. It features a large 'Table: Appointments' and an 'Add Appointment' button in the top right corner. Below the table, there is a 'Buttons: View Appt.' button.

Figure 2.6.5: Display Appointments Page

Figure 2.6.5 shows the Appointment Management Page. Users can view appointment details and monitor booking statuses.

Tab

URL box

Navigation bar

Book Appointment

Form: Book Appointment

Counselor 1

Counselor 2

Counselor 1 information

Counselor 2 information

Time: Dropdown: Times

Time: Dropdown: Times

Button: Book Appt.

Button: Book Appt.

Button: Feedback

Button: Profile

Button: Logout

Figure 2.6.6.: Book Session Page

Figure 2.6.6 shows the Book Session page. Students can select available counselling slots and submit appointment requests.

Tab

URL box

Navigation bar

Feedback

Form: Student Feedback

Table: Counseling Sessions

Text box: Student Feedback

Button: Save Feedback

Button: Book Appt.

Button: Feedback

Button: Profile

Button: Logout

Figure 2.6.7: Feedback Page

Figure 2.6.7 shows the Student Feedback Form. Students can provide ratings and comments after completing counselling sessions.

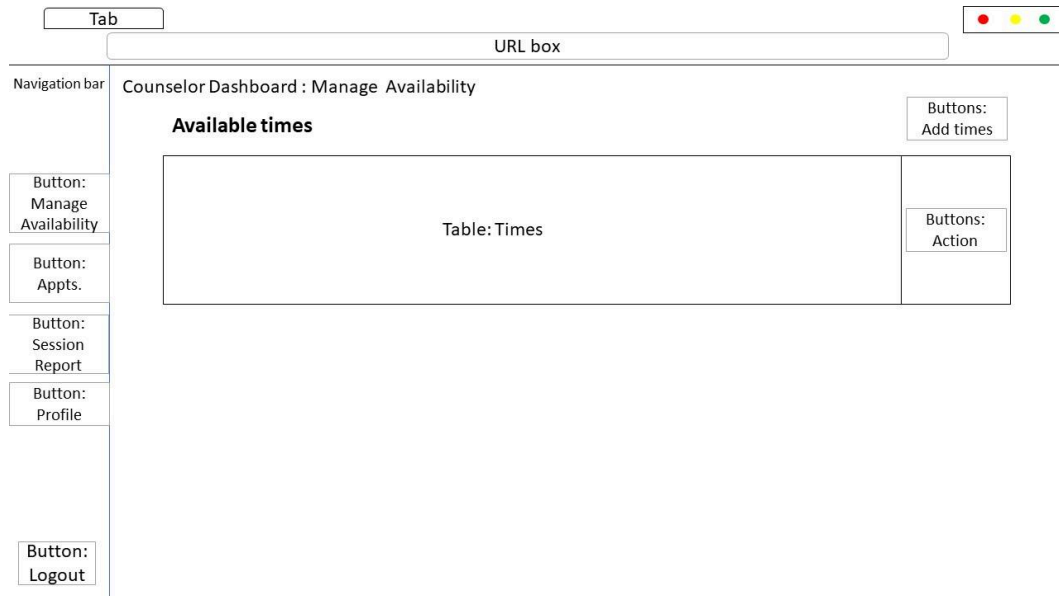


Figure 2.6.8: Manage Availability Page

Figure 2.6.8 shows the Manage Availability page. Counsellors can manage their available dates and time slots.

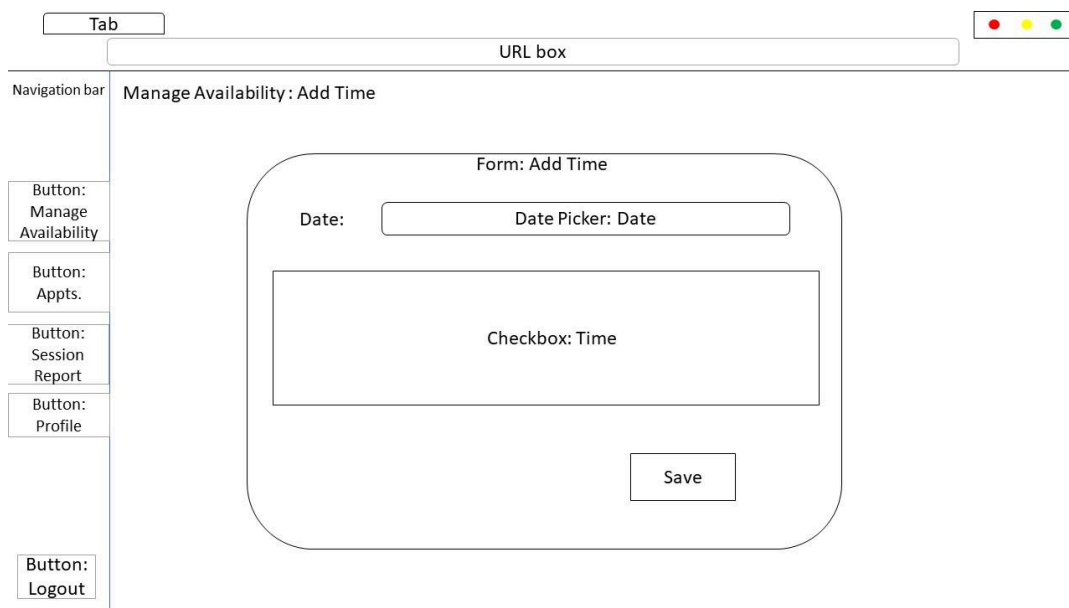


Figure 2.6.9 : Add Availability Page

Figure 2.6.9 shows the Add Availability interface. Counsellors can create new availability schedules for future counselling sessions.

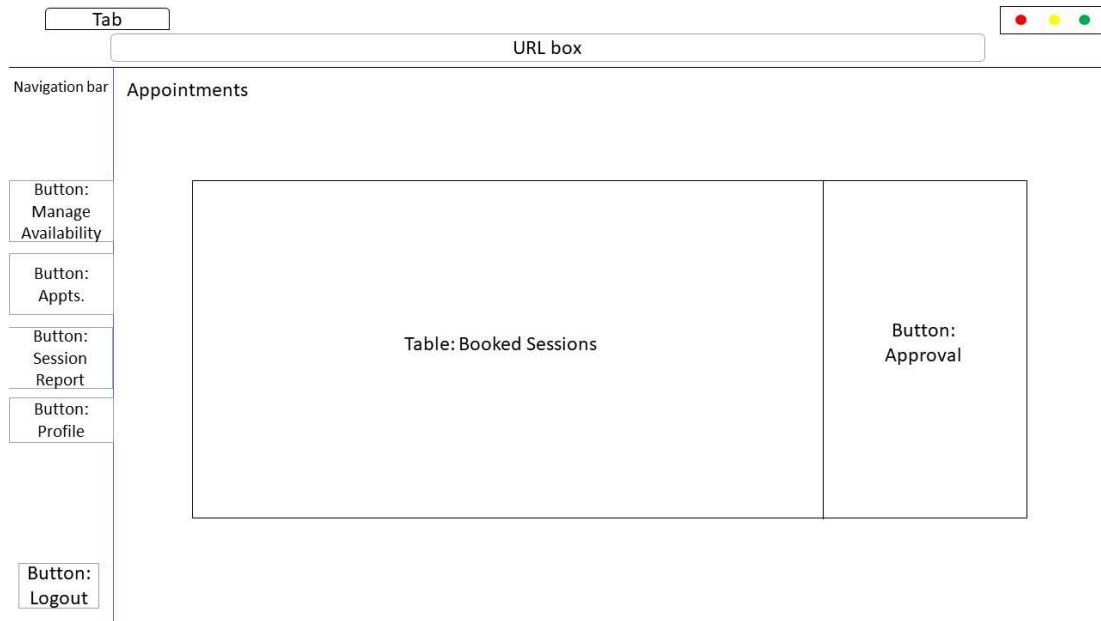


Figure 2.6.10 : Booked Sessions Page

Figure 2.6.10 shows the Booked Sessions page. Counsellors can review appointment requests and manage booking approvals.



Figure 2.6.11: Session Records Page

Figure 2.6.11 shows the Session Records page. Counsellors can record and review counselling session notes and reports.

The image shows a web application interface for editing a profile. At the top, there is a 'Tab' and a 'URL box'. Below this is a 'Navigation bar' with the text 'Edit Profile Page'. The main content area contains a 'Form: Edit Profile' with four text input fields: 'Name', 'ID', 'Phone Number', and 'Password'. Each field has a label and a corresponding text box. A 'Save' button is located at the bottom right of the form. On the left side of the page, there is a 'Button: Logout'.

Tab

URL box

Navigation bar

Edit Profile Page

Form: Edit Profile

Name: Text Box: Name

ID: Text Box: ID Number

Phone Number: Text Box: Phone Number

Password: Text Box: Password

Save

Button: Logout

Figure 2.6.12: Edit Profile Page

Figure 2.6.12 shows the Edit Profile page. Users can update personal information and maintain their account details.

2.7 Description of major pages and functionalities

1. **Login Page:** Authorizes users to use the system. The system redirects users to relevant dashboard pages based on user type: Admin, Counselor and Student.
2. **Display Student Page:** Displays all students. Only can edit students' profiles like phone number but cannot edit for student name, email, studentID. Cannot add or delete students, student data are related to university student database, power of CRUD should be limited to make sure the alignment of data.
3. **Display Counselor Page:** Displays all counselors. Has functions to add, edit and delete counselor accounts.
4. **Appointment booking page:** Displays list of available counselors and available sessions. Students can pick available slots from the list, as well as reschedule appointments.
5. **Student Feedback Form:** After counseling session, student may leave a satisfaction review of the session.
6. **Counselor Availability Page:** Counselor can add time slots for days and times where they can meet students.
7. **Appointments Page:** Counselor can view appointments booked by students and approve the time slots.
8. **Session Review Page:** Counselor can write a session report after the counseling session is over
9. **Profile Editing Page:** Counselors and students may edit their personal information and passwords.

3.0 System Implementation

3.1 Development tools used

Tool	Purpose
NetBeans IDE	Used to develop Java classes, Servlets, JSP pages and project configuration.
Apache Tomcat	Used as the web application server.
XAMPP	Used to provide the MySQL database environment.
phpMyAdmin	Used to create, import, update and test database records.
Web Browser	Used to test pages, forms, validation and system workflow.

Table 3.1: Development Tools Used

Table 3.1 shows the development tools used throughout the implementation of the Counseling Session Booking System. These tools support coding, database management, testing and deployment activities.

3.2 Technologies used

Technology	Purpose
Java	Main programming language for business logic.
Java Servlet	Handles HTTP requests, responses, user roles, validation and CRUD processes.
JSP	Displays dynamic web pages using data retrieved from the database.
MySQL	Stores users, counsellors, schedules, bookings and session records.
JavaBeans	Represents entities such as User, Counsellor, Schedule, Booking and SessionRecord.
DAO Classes	Manage SQL queries and database communication.

HTML and CSS	Create the structure, layout, forms, tables, cards and buttons.
JavaScript	Performs client-side input validation and interactive schedule checks.

Table 3.2: Technologies Used

Table 3.2 shows the technologies used in the development of the system. Each technology contributes to different aspects such as front-end development, server-side processing, database management and application architecture.

3.3 Description of Implemented Features

3.3.1 Login, Student List and Profile Management

- Users log in using a User ID and password. The User ID is converted to uppercase before login validation.
- Student accounts are inserted as dummy data in the database. Students cannot self-register through the system. The administrator can view the Student List and update student phone numbers only. Student ID, full name, email and role are read-only.
- All roles can access My Profile and Change Password functions.

3.3.2 Counsellor Management

Counsellor Management supports complete CRUD operations.

CRUD Operation	Implementation
Create	Administrator adds a new counsellor account and profile.
Retrieve	Administrator views the counsellor list.
Update	Administrator updates counsellor name, specialization, email, phone number and office location.
Delete	Administrator deletes a counsellor account.

Table 3.3.2 Counsellor Management Table

Table 3.3.2 shows Counsellor Management Table. The Counsellor ID is generated automatically from the email prefix. For example, c001@counselling.edu.my becomes C001.

3.3.3 Schedule Management

Counsellors can create preset one-hour slots and one custom one-hour slot. The system displays available slots in a weekly card format and a selected-day detail page.

CRUD Operation	Implementation
Create	Counsellor creates preset or custom one-hour availability slots.
Retrieve	Counsellor views weekly availability and selected-day slot details.
Update	Counsellor updates an unused availability slot.
Delete	Counsellor deletes an unused availability slot.

Table 3.3.3 Schedule Management Table

Table 3.3.3 shows Schedule Management Table. The system prevents duplicate or overlapping slots. Booked slots cannot be edited or deleted.

3.3.4 Booking Management

Students can view available counselling slots and create booking requests. The booking is initially saved with a PENDING status.

CRUD Operation	Implementation
Create	The student selects an available slot and submits a booking request.
Retrieve	Students, Counsellors and Administrators view booking lists according to their role.
Update	Students reschedule pending bookings. Counsellors approve or reject pending bookings.

Delete	Students, Counsellors or Administrators cancel bookings logically by changing booking status to CANCELLED.
--------	--

Table 3.3.4 Booking Management Table

Table 3.3.4 shows the Booking Management Table. The booking page shows counsellor name, specialization, date, time and office location to help students select suitable sessions.

3.3.5 Session Record and Feedback Management

Counsellors can record session notes after a counselling session. Students can view completed sessions and submit feedback.

CRUD Operation	Implementation
Create	Counselor creates session notes. Student submits feedback and rating.
Retrieve	Counsellor views session records. Student views session history.
Update	Counsellor edits session notes. Student can edit feedback after a session record is saved (meaning a counseling session ends).
Delete	Counsellor deletes session records when necessary.

Table 3.3.5 Session Record and Feedback Management Table

Table 3.3.5 shows Session Record and Feedback Management Table. Counselors can add session records. After the counselor saves session notes. The counseling session ends and students can give feedback to the corresponding counseling session.

3.3.6 Input Validation

Our system uses JavaScript validation to reduce invalid input before forms are submitted.

Validation includes:

- Required field validation
- Email format validation
- Malaysian phone-number validation

- Password validation
- Password confirmation validation
- Rating validation from 1 to 5
- Future-date validation
- Schedule-overlap validation

Besides, we also have Server-side validation which is performed in Servlets for user role access, booking actions, phone-number format, password format and schedule-overlap control.

3.4 Screenshots of the developed web pages

3.4.1 System Login Page

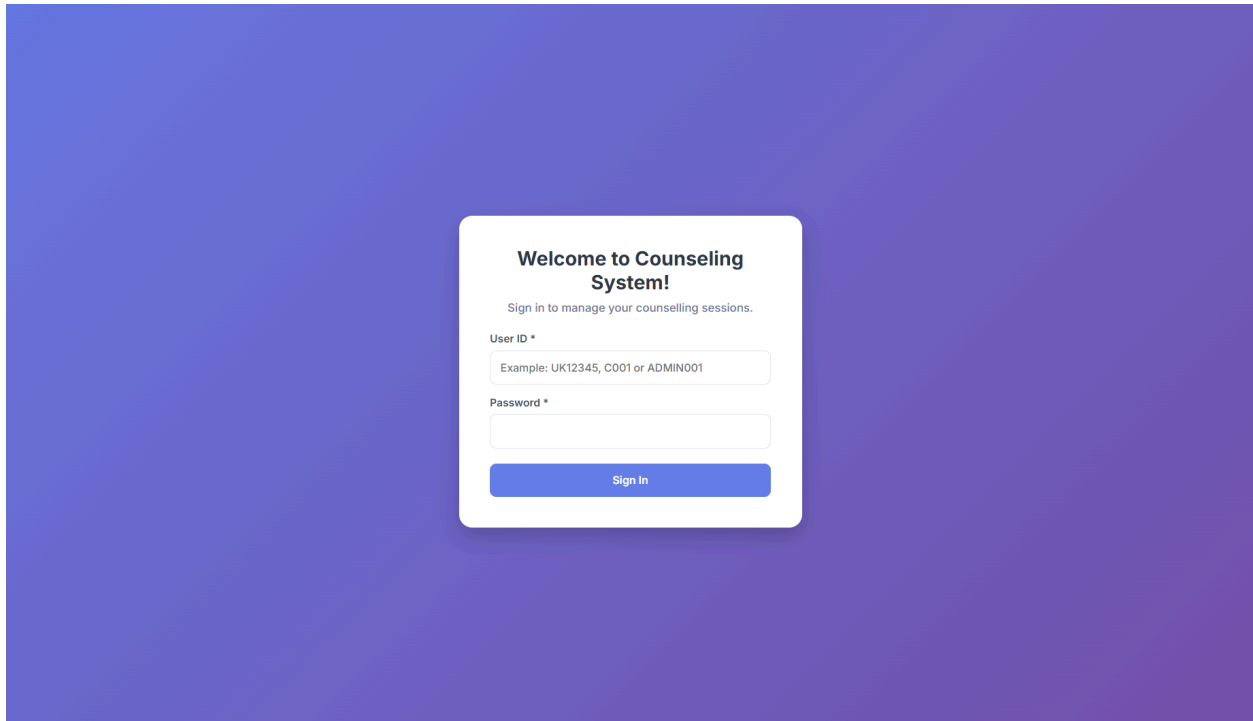


Figure 3.4.1: System Login Page

Figure 3.4.1 shows the centralized login interface of the Counselling Session Booking System. The page features a clean, user-friendly design where users must input their specific User ID and Password to gain access. This portal handles role-based authentication for all system actors. It accepts credentials for students (e.g. UK12345), counsellors (e.g. C001) and administrators (e.g. ADMIN001) and automatically routes them to their corresponding dashboards if verification is successful.

3.4.2 Admin Dashboard

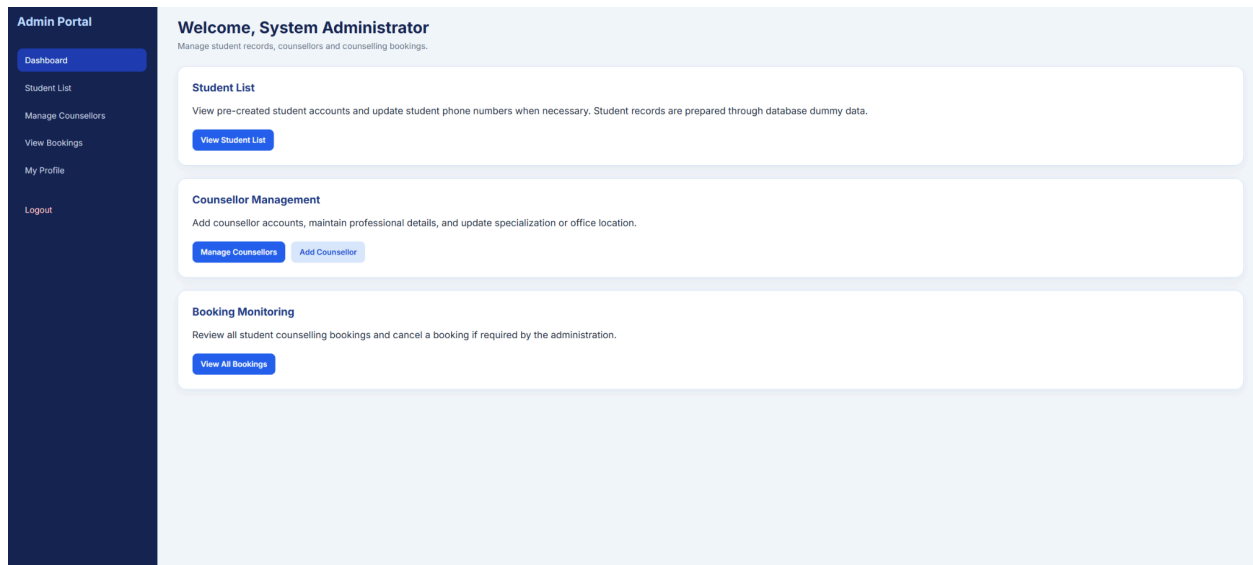


Figure 3.4.2: Admin Dashboard

Figure 3.4.2 displays the Admin Dashboard which serves as the central control panel for the System Administrator. The left sidebar provides quick navigation to various management modules including Student List, Manage Counsellors, View Bookings and My Profile. The main workspace features dedicated, easy-to-read sections for "Student List", "Counsellor Management" and "Booking Monitoring". Through this interface, the administrator can easily navigate to perform essential operations such as viewing pre-created student records, adding or updating counsellor details and reviewing or cancelling session bookings.

3.4.3 Admin Student List

Student List					
View existing student accounts and update phone numbers when necessary. Student accounts are prepared through database dummy data.					
Student Accounts					
No.	Student ID	Full Name	Email	Phone Number	Actions
1	ST2345	Daniel Tan Wei Ming	st2345@gmail.com	011-23456787	Edit Phone
2	UK12345	Nur Aina Rahman	uk12345@gmail.com	012-3456789	Edit Phone
3	UK12346	Siti Farah Hassan	uk12346@gmail.com	013-5678901	Edit Phone

Figure 3.4.3: Admin Student List Page

Figure 3.4.3 displays the Student List page within the Admin Portal. This interface presents a clear, tabular view of all existing student accounts in the system. The organized table displays key user information, including the Student ID, Full Name, Email and Phone Number. Additionally, an "Actions" column provides an "Edit Phone" button, so this gives the administrator the capability to quickly update a student's contact number directly from the list.

3.4.4 Edit Student Phone Number Page

The screenshot displays the 'Edit Student Phone Number' page within an Admin Portal. The sidebar on the left contains the following navigation links: Dashboard, Student List (highlighted), Manage Counsellors, View Bookings, My Profile, and Logout. The main content area is titled 'Edit Student Phone Number' and includes a note: 'Student account details are maintained through the university database. Only the phone number can be updated in this system.' The form contains four input fields: 'Student ID' with the value 'S12345', 'Full Name' with 'Daniel Tan Wei Ming', 'Email' with 's12345@gmail.com', and 'Phone Number' with '011-23456787'. At the bottom of the form are two buttons: 'Update Phone Number' and 'Cancel'.

Field	Value	Editable
Student ID	S12345	No
Full Name	Daniel Tan Wei Ming	No
Email	s12345@gmail.com	No
Phone Number	011-23456787	Yes

Figure 3.4.4: Edit Student Phone Number Interface

Figure 3.4.4 shows the "Edit Student Phone Number" form which is accessed from the Admin Student List. To maintain data integrity with the university's primary database, core account details such as the Student ID, Full Name and Email are displayed as read-only fields. The administrator is only permitted to modify the student's Phone Number. This restricted editing capability ensures contact details can be kept up to date without risking accidental changes to official student records. The form includes an "Update Phone Number" button to save the new contact information and a "Cancel" button to return to the list without making changes.

3.4.5 Manage Counsellors Page

Admin Portal

- Dashboard
- Student List
- Manage Counsellors**
- View Bookings
- My Profile
- Logout

Manage Counsellors

Creates, updates, and removes counsellor accounts. The Counsellor ID is generated automatically from the email text before the @ symbol.

[+ Add Counsellor](#)

No.	Counsellor ID	Full Name	Specialization	Email	Phone Number	Office Location	Actions
1	C001	Dr. Sarah Smith	Academic Stress and Study Skills	c001@counselling.edu.my	012-4567890	Student Wellbeing Centre, Room C-203	Edit Delete
2	C002	Dr. John Lee	Personal and Career Counselling	c002@counselling.edu.my	011-34567890	Student Wellbeing Centre, Room C-205	Edit Delete
3	C003	Dr. Syamil Ahmad	Student Mental Health	c003@counselling.edu.my	013-4567897	Student Wellbeing Centre, Room C-207	Edit Delete

Figure 3.4.5: Manage Counsellors Page

Figure 3.4.5 displays the Manage Counsellors page within the Admin Portal. This interface allows the system administrator to oversee the counseling staff by displaying a comprehensive list of counsellor profiles. The data table includes professional details such as the Counsellor ID, Full Name, Specialization, Email, Phone Number and Office Location. From this page, the administrator can perform complete management operations: creating new accounts via the "+ Add Counsellor" button at the top right or using the "Edit" and "Delete" action buttons located on each row to update or remove existing records. Additionally, the system provides a helpful prompt noting that the Counsellor ID is automatically generated from the provided email address.

3.4.6 Add Counsellor Page

The screenshot shows the 'Add Counsellor' form within the Admin Portal. The left sidebar contains links for Dashboard, Student List, Manage Counsellors (highlighted), View Bookings, My Profile, and Logout. The main content area has the title 'Add Counsellor' and a subtitle 'Create a counsellor account and profile. The Counsellor ID is generated from the email address.' The form includes the following fields: 'Counsellor ID' (with a note 'Generated from email'), 'Full Name *', 'Email *', 'Phone Number' (with a placeholder '012-3456789'), 'Password *', 'Specialization *', and 'Office Location'. At the bottom are 'Create Counsellor' and 'Cancel' buttons.

Figure 3.4.6: Add Counsellor Form

Figure 3.4.6 illustrates the form utilized by the administrator to create a new counsellor profile. This page requires mandatory input fields including the counsellor's Full Name, Email, Password and Specialization. To facilitate efficient data entry, the Counsellor ID field is set to automatically generate based on the email provided. Additionally, the administrator can optionally provide the counsellor's Phone Number and Office Location to complete the professional profile which is then submitted via the "Create Counsellor" button.

3.4.7 Edit Counsellor Page

The screenshot shows the 'Edit Counsellor' form within the Admin Portal. The left sidebar is identical to the previous figure. The main content area has the title 'Edit Counsellor' and a subtitle 'Update the counsellor profile. The Counsellor ID follows the email address.' The form includes the following fields: 'Counsellor ID' (with a value 'C001'), 'Full Name *' (with a value 'Dr. Sarah Smith'), 'Email *' (with a value 'c001@counselling.edu.my'), 'Phone Number' (with a value '012-4567890'), 'Specialization *' (with a value 'Academic Stress and Study Skills'), and 'Office Location' (with a value 'Student Wellbeing Centre, Room C-203'). A note states 'Password changes are handled separately.' At the bottom are 'Update Counsellor' and 'Cancel' buttons.

Figure 3.4.7: Edit Counsellor Form

Figure 3.4.7 illustrates the "Edit Counsellor" form which allows the administrator to update existing staff information. To ensure consistency within the system, the Counsellor ID field is set to read-only. All other profile details, including the Full Name, Email, Phone Number,

Specialization and Office Location are fully editable. This allows the administrator to reflect changes in a counsellor's professional details or contact information. The page also provides "Update Counsellor" and "Cancel" buttons to finalize or discard changes.

3.4.8 Booking Monitoring Page

Admin Portal

Dashboard

Student List

Manage Counsellors

View Bookings

My Profile

Logout

Booking Monitoring

View all counselling bookings across the system.

All Counselling Bookings

No.	Student	Counsellor	Date	Time	Booking Date	Status	Action
1	Nur Aisha Rahman	Dr. Sarah Smith	2026-06-27	10:00:00	2026-06-25	Pending	Cancel Booking
2	Siti Farah Hassan	Dr. John Lee	2026-06-23	14:00:00	2026-06-21	Completed	-
3	Nur Aisha Rahman	Dr. John Lee	2026-06-27	10:00:00	2026-06-24	Rejected	-
4	Daniel Tan Wei Ming	Dr. Sarah Smith	2026-06-27	14:00:00	2026-06-24	Cancelled by Student	-
5	Siti Farah Hassan	Dr. John Lee	2026-06-26	11:00:00	2026-06-25	Approved	Cancel Booking
6	Daniel Tan Wei Ming	Dr. Sarah Smith	2026-06-26	10:00:00	2026-06-25	Approved	Cancel Booking
7	Nur Aisha Rahman	Dr. Sarah Smith	2026-06-22	10:00:00	2026-06-20	Completed	-

Figure 3.4.8: Booking Monitoring Interface

Figure 3.4.8 displays the Booking Monitoring interface within the Admin Portal. This comprehensive table provides the administrator with oversight of all counselling appointments across the system. It lists essential details for each entry, including the student name, assigned counsellor, appointment date and time, the booking date and the current booking status (e.g. Pending, Completed, Rejected, Approved or Cancelled by Student). The interface also includes an "Action" column which allows the administrator to perform administrative power such as cancelling specific bookings when necessary.

3.4.9 My Profile Page

The screenshot shows the 'My Profile' page within the 'Admin Portal'. The left sidebar contains links: Dashboard, Student List, Manage Counsellors, View Bookings, My Profile (highlighted), Change Password, and Logout. The main content area is titled 'My Profile' with a subtitle 'Update your personal account details.' Below this is a form titled 'Account Information'. It contains five input fields: 'Username' (pre-filled with 'ADMIN001'), 'Account Role' (pre-filled with 'Administrator'), 'Full Name *' (pre-filled with 'System Administrator'), 'Email *' (pre-filled with 'admin@counselling.edu.my'), and 'Phone Number *' (pre-filled with '012-12356723'). At the bottom of the form are three buttons: 'Save Changes' (in blue), 'Change Password' (in light blue), and 'Cancel' (in light blue).

Figure 3.4.9: My Profile Interface

Figure 3.4.9 displays the "My Profile" page which allows the currently logged-in user to view and update their personal account details. For the purpose of data integrity, the Username and Account Role fields are set as read-only. The system is designed to allow the user to modify their Full Name, Email and Phone Number. Besides, the profile structure is consistent across the Admin, Counsellor and Student portals to ensure a uniform user experience. Users can apply changes using the "Save Changes" button, initiate a security update via the "Change Password" button or exit the view using "Cancel".

3.4.10 Change Password Page

The screenshot shows the 'Change Password' page within the 'Admin Portal'. The left sidebar is identical to the previous figure, with 'Change Password' highlighted. The main content area is titled 'Change Password' with a subtitle 'Use at least 6 characters, including one letter and one number. Spaces are not allowed.' Below this is a form titled 'Password Information'. It contains three input fields: 'Current Password *', 'New Password *', and 'Confirm New Password *'. At the bottom of the form are two buttons: 'Update Password' (in blue) and 'Cancel' (in light blue).

Figure 3.4.10: Change Password Interface

Figure 3.4.10 illustrates the "Change Password" interface which is available to all system users to maintain account security. To prevent unauthorized password modifications, the page requires the user to enter their current password for validation, then followed by a new password and a confirmation field. The system enforces password complexity requirements, explicitly

instructing users to include at least 6 characters, at least one letter and one number. This is just a simple security password restriction.

3.4.11 Counsellor Dashboard

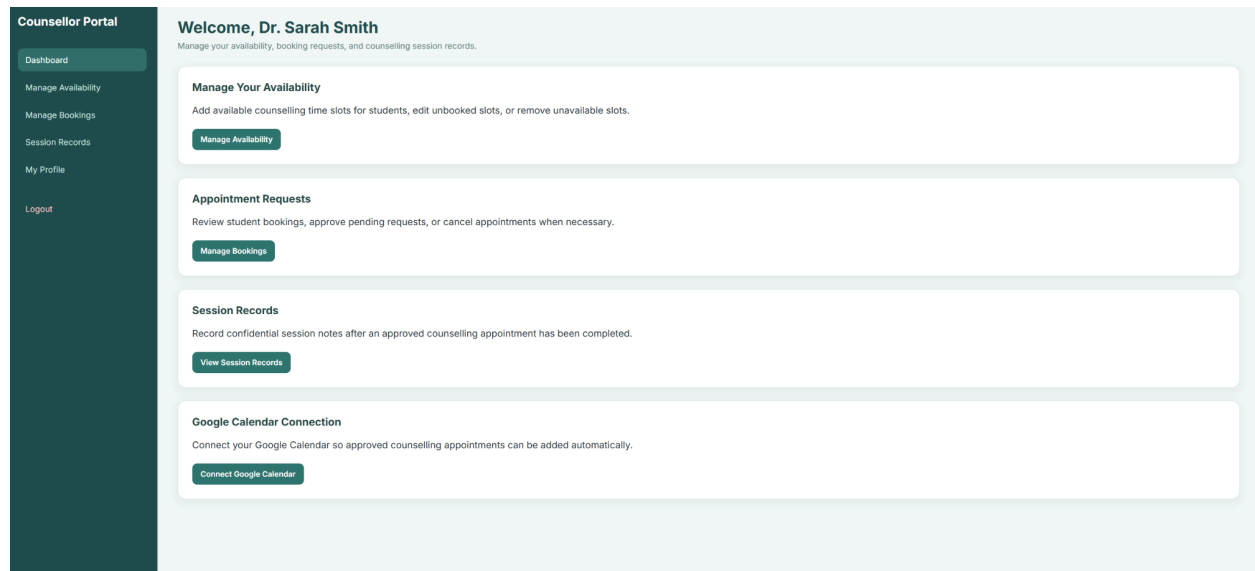


Figure 3.4.11: Counsellor Dashboard

Figure 3.4.11 illustrates the Counsellor Dashboard which provides a centralized control hub for Counselor. The interface offers quick access to essential management modules including "Manage Your Availability", "Appointment Requests", "Session Records" and "Google Calendar Connection." Each section includes a descriptive summary of its functionality and a direct call-to-action button make it very user friendly.

3.4.12 Manage Availability

The screenshot shows the 'Manage Availability' page in the Counsellor Portal. The left sidebar contains links for Dashboard, Manage Availability (active), Manage Bookings, Session Records, My Profile, and Logout. The main content area is titled 'Manage Availability' and includes a sub-header 'View your availability for today and the next six days.' and a '+ Add Availability' button. Below this is a 'Weekly Availability' section displaying a grid of seven cards for each day of the week (Today, Fri, Sat, Sun, Mon, Tue, Wed). Each card shows the date, the number of slots, and a list of time slots. Slots are either 'AVAILABLE' (light green) or 'BOOKED' (orange). For example, 'Today' (25 Jun) has 2 slots: 09:00 AM - 10:00 AM (AVAILABLE) and 11:00 AM - 12:00 PM (AVAILABLE). 'Fri' (26 Jun) has 3 slots: 09:00 AM - 10:00 AM (AVAILABLE), 10:00 AM - 11:00 AM (BOOKED), and 02:00 PM - 03:00 PM (AVAILABLE). 'Sat' (27 Jun) has 2 slots: 10:00 AM - 11:00 AM (BOOKED) and 02:00 PM - 03:00 PM (AVAILABLE). 'Sun' (28 Jun) has 2 slots: 09:00 AM - 10:00 AM (AVAILABLE) and 03:00 PM - 04:00 PM (AVAILABLE). 'Mon' (29 Jun) has 0 slots. 'Tue' (30 Jun) has 1 slot: 10:00 AM - 11:00 AM (AVAILABLE). 'Wed' (01 Jul) has 1 slot: 02:00 PM - 03:00 PM (AVAILABLE). Each card has a 'Details' button at the bottom.

Figure 3.4.12: Manage Availability Interface

Figure 3.4.12 displays the Manage Availability page. This view presents a weekly calendar grid which allows the counsellor to visualize their scheduled slots for the current day and the upcoming six days. The interface clearly distinguishes between "AVAILABLE" and "BOOKED" slots and provides a "Details" option for further management of specific time slots. Users can also utilize the "+ Add Availability" button to quickly add new availability.

3.4.13 Add Availability

The screenshot shows the 'Add Availability' form in the Counsellor Portal. The left sidebar is the same as in Figure 3.4.12. The main content area is titled 'Add Availability' and includes a sub-header 'Create preset slots and one optional custom one-hour slot.' Below this is a form with the following fields: 'Available Date *' (a dropdown menu with '-- Select a Date --'), 'Preset One-Hour Slots' (a section with the instruction 'Choose common slots, then optionally add one custom one-hour slot below.' and a 'Select All' button), and 'Custom One-Hour Slot (Optional)' (a section with a time range input field and a 'Create Availability' button). The 'Preset One-Hour Slots' section contains six checkboxes for the following time slots: 09:00 AM - 10:00 AM, 10:00 AM - 11:00 AM, 11:00 AM - 12:00 PM, 02:00 PM - 03:00 PM, 03:00 PM - 04:00 PM, and 04:00 PM - 05:00 PM. The 'Custom One-Hour Slot (Optional)' section contains a time range input field with a placeholder '--:-- --' and a 'Create Availability' button. Below the form is a note: 'Example: choosing 10:30 AM creates 10:30 AM - 11:30 AM. A slot is automatically available for booking after it is created. Booked slots cannot be edited or deleted.'

Figure 3.4.13: Add Availability Form

Figure 3.4.13 shows the Add Availability form which allows the counsellor to create new appointment slots. The form requires the selection of an "Available Date" from a dropdown menu, then followed by a selection of preset one-hour slots using checkboxes. An optional

"Custom One-Hour Slot" field is provided for flexible scheduling. Once configured, the "Create Availability" button generates these slots in the database to make it immediately available for student booking.

3.4.14 Manage Bookings

Counsellor Portal

Dashboard

Manage Availability

Manage Bookings

Session Records

My Profile

Logout

Manage Bookings

Review active bookings and completed booking history.

Student Bookings

No.	Student	Email	Phone	Date	Time	Status	Actions
1	Nur Aina Rahman	uk12345@gmail.com	012-3456789	2026-06-22	10:00:00	COMPLETED	-
2	Daniel Tan Wei Ming	s12345@gmail.com	011-23456787	2026-06-26	10:00:00	APPROVED	Record Session Cancel
3	Nur Aina Rahman	uk12345@gmail.com	012-3456789	2026-06-27	10:00:00	PENDING	Approve Reject
4	Daniel Tan Wei Ming	s12345@gmail.com	011-23456787	2026-06-27	14:00:00	CANCELLED BY STUDENT	-

Figure 3.4.14: Manage Bookings Table

Figure 3.4.14 provides the Manage Bookings interface, used for reviewing active bookings and historical session data. The table tracks bookings by Student Name, Email, Phone Number, Date, Time and Status. The "Action" column includes buttons such as "Record Session" for completed appointments and "Approve" or "Reject" buttons for handling pending student requests.

3.4.15 Session Records

Counsellor Portal

Dashboard

Manage Availability

Manage Bookings

Session Records

My Profile

Logout

Session Records

Review, update, or remove completed counselling session notes.

View Approved Bookings

Completed Session Records

No.	Student	Session Date	Session Time	Notes	Feedback	Rating	Actions
1	Nur Aina Rahman	2026-06-22	10:00:00	The student discussed academic stress and difficulty managing study time. A weekly study plan and follow-up actions were agreed.	The counsellor was supportive and the study plan was practical.	5 / 5	Edit Notes Delete

Figure 3.4.15: Session Records Table

Figure 3.4.15 displays the Session Records table which lists all completed counselling sessions. The table includes specific details for each record such as the student's name, the date and time of the session, the confidential counselling notes recorded by the counsellor, student feedback

and the provided session rating. The "Actions" column allows the counsellor to "Edit Notes" or "Delete" records as required.

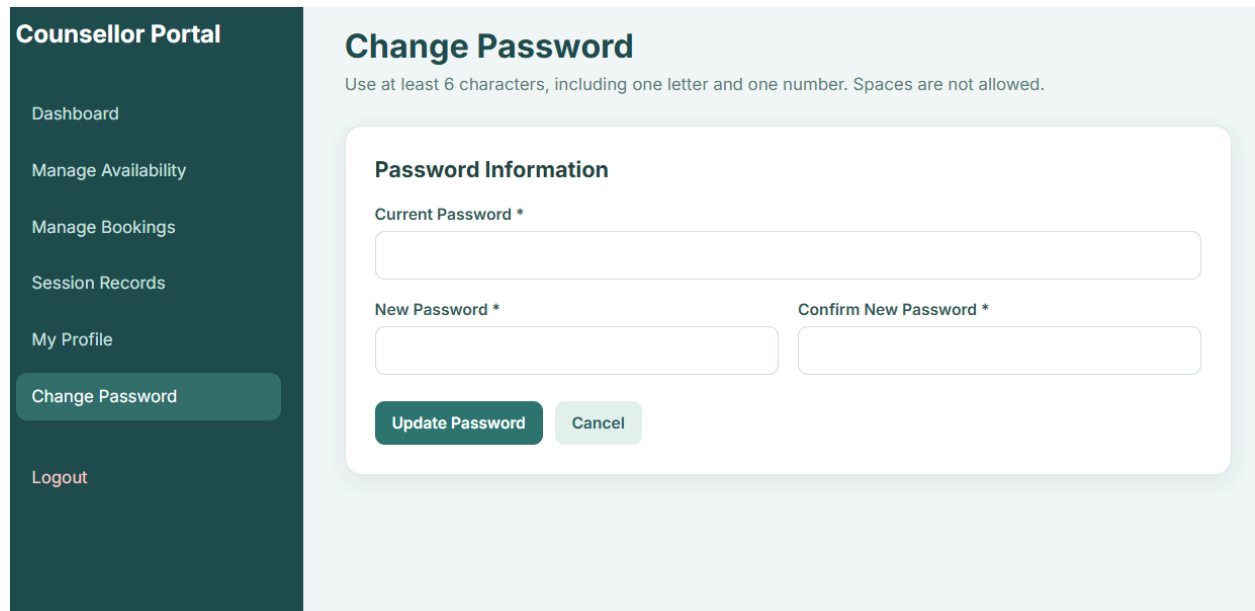
3.4.16 Counsellor Profile

The screenshot shows the 'Counsellor Portal' interface. On the left is a dark green sidebar with a list of navigation items: 'Dashboard', 'Manage Availability', 'Manage Bookings', 'Session Records', 'My Profile' (which is highlighted), 'Change Password', and 'Logout'. The main content area is titled 'My Profile' with a subtitle 'Update your personal account details.' Below this is a form titled 'Account Information'. The form contains five input fields: 'Username' (with value 'C001'), 'Account Role' (with value 'Counsellor'), 'Full Name *' (with value 'Dr. Sarah Smith'), 'Email *' (with value 'c001@counselling.edu.my'), and 'Phone Number *' (with value '012-4567890'). At the bottom of the form are three buttons: 'Save Changes' (in green), 'Change Password' (in light green), and 'Cancel' (in light green).

Figure 3.4.16: Counsellor My Profile Interface

Figure 3.4.16 presents the "My Profile" page for the counsellor. Consistent with other portal profiles, the Username and Account Role fields are read-only to maintain system identity. The counsellor may update their Full Name, Email and Phone Number as necessary. The page provides a clear interface for saving changes and a button for changing password or cancel button to exit change.

3.4.17 Counsellor Change Password



Counsellor Portal

- Dashboard
- Manage Availability
- Manage Bookings
- Session Records
- My Profile
- Change Password**
- Logout

Change Password

Use at least 6 characters, including one letter and one number. Spaces are not allowed.

Password Information

Current Password *

New Password *

Confirm New Password *

Update Password Cancel

Figure 3.4.17: Counsellor Change Password Interface

Figure 3.4.17 shows the "Change Password" interface for the counsellor portal. This form have the same standard as the other role's change password portals. It needs a "Current Password" for verification and a "New Password" and "Confirm New Password" to change password. All new passwords must include a minimum length of 6 characters containing at least one letter and one number with no spaces.

3.4.18 Student Dashboard

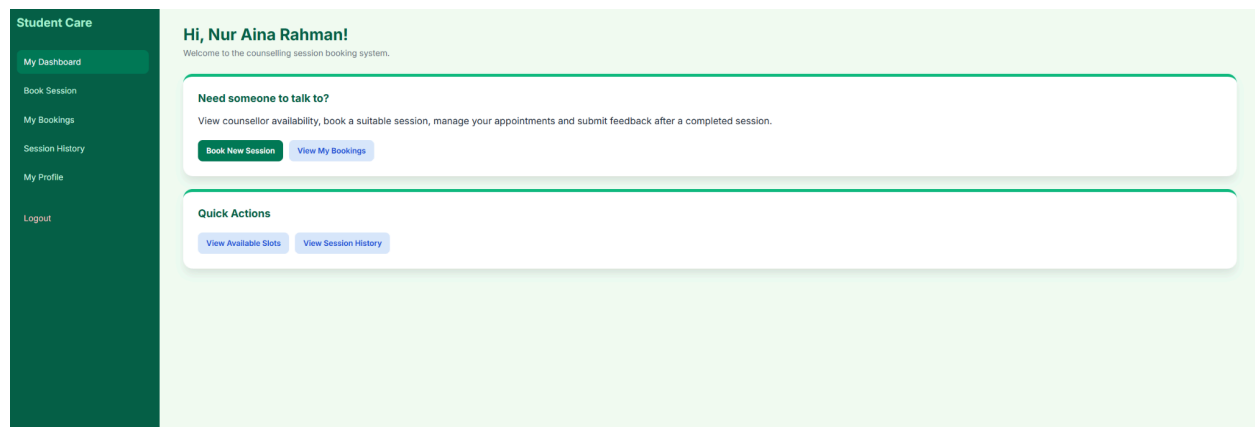


Figure 3.4.18: Student Dashboard

Figure 3.4.18 illustrates the Student Dashboard. This page serves as the main entry point for students. It provides a clear overview of their "Student Information" (Name, ID and Phone Number) and a section for "Booked Appointments." The sidebar provides quick navigation buttons to "Book Appointment", "Session History" and "Feedback Form".

3.4.19 Book Appointment

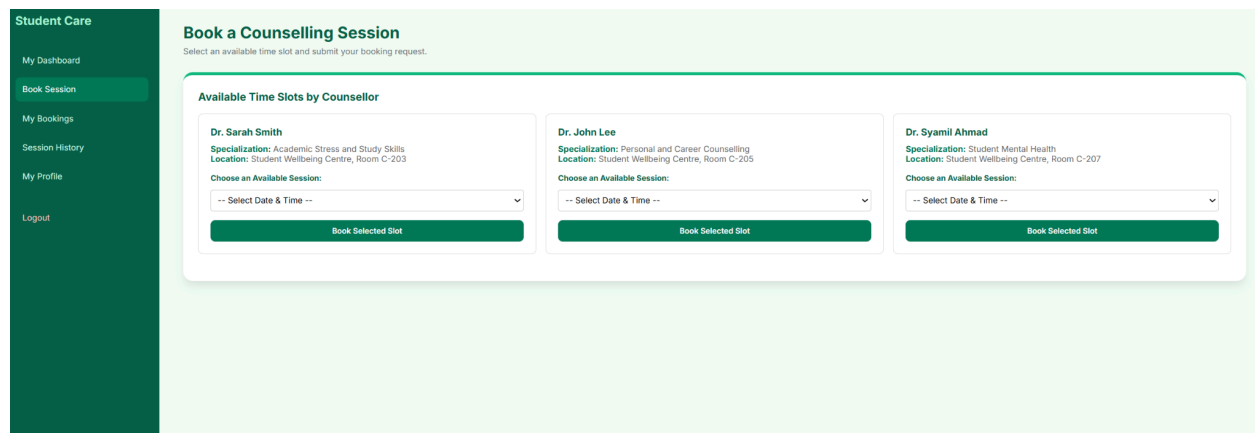
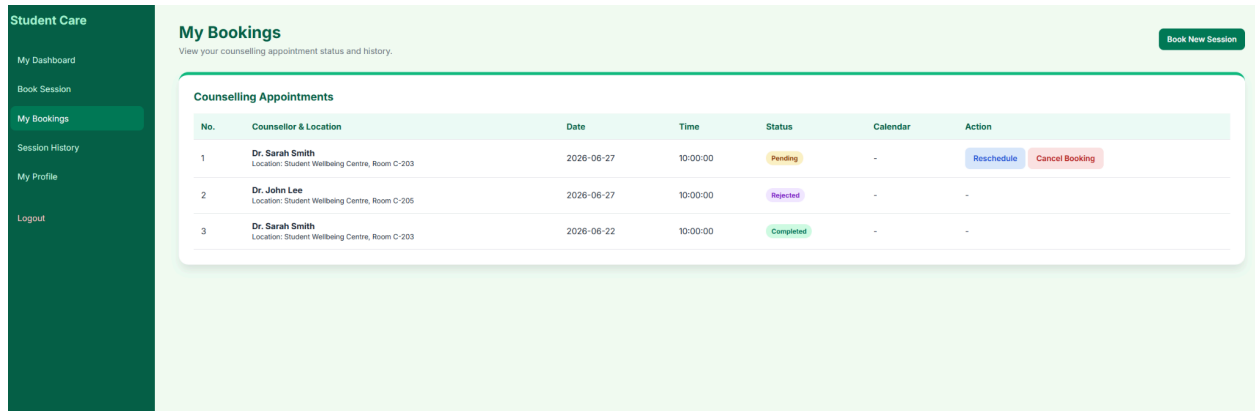


Figure 3.4.19: Book Appointment Interface

Figure 3.4.19 displays the Book Appointment interface. Students can select their preferred "Counsellor", "Date" and "Time" from the provided dropdown and date-picker menus. Once the desired slot is selected, students click the "Book Selected Slot" button to confirm their appointment which automatically updates the system's schedule status in the database to prevent double booking.

3.4.20 Student Session History



The screenshot shows a web interface for 'My Bookings'. On the left is a dark green sidebar with navigation links: 'Student Care', 'My Dashboard', 'Book Session', 'My Bookings' (highlighted), 'Session History', 'My Profile', and 'Logout'. The main content area has a light green background. At the top, it says 'My Bookings' with a subtitle 'View your counselling appointment status and history.' and a 'Book New Session' button. Below this is a table titled 'Counselling Appointments' with columns: No., Counsellor & Location, Date, Time, Status, Calendar, and Action. The table contains three rows of data.

No.	Counsellor & Location	Date	Time	Status	Calendar	Action
1	Dr. Sarah Smith Location: Student Wellbeing Centre, Room C-203	2026-06-27	10:00:00	Pending	-	Reschedule Cancel Booking
2	Dr. John Lee Location: Student Wellbeing Centre, Room C-205	2026-06-27	10:00:00	Rejected	-	-
3	Dr. Sarah Smith Location: Student Wellbeing Centre, Room C-203	2026-06-22	10:00:00	Completed	-	-

Figure 3.4.20: Session History Table

Figure 3.4.20 shows the Student Session History page which allows students to review their past appointments. The table displays the "Counsellor Name", "Date", "Time", "Status" and provides an "Action" column where students can view details of past sessions or provide feedback for a completed counselling session.

3.4.21 Student Session History

The screenshot shows a web application interface for 'Student Care'. On the left is a dark green sidebar with a 'Student Care' header and a list of links: 'My Dashboard', 'Book Session', 'My Bookings', 'Session History' (highlighted in green), 'My Profile', and 'Logout'. The main content area has a light green background. At the top, it says 'Session History' with a subtitle 'View completed counselling sessions and submit feedback.' Below this is a white box titled 'Completed Sessions' containing a table. The table has columns for 'No.', 'Counsellor', 'Date', 'Time', 'Feedback', 'Rating', and 'Action'. It contains one row for session 1, with a rating of 5/5 and a 'View / Edit Feedback' button.

No.	Counsellor	Date	Time	Feedback	Rating	Action
1	Dr. Sarah Smith	2026-06-22	10:00:00	The counsellor was supportive and the study plan was practical.	5 / 5	View / Edit Feedback

Figure 3.4.21: Reschedule Booking Form

Figure 3.4.21 shows a Student Session History. This interface allows students to modify his feedback or view his feedback.

3.4.22 Feedback Form

The screenshot shows the 'Session Feedback' form. The sidebar is identical to the previous figure. The main content area has a light green background. At the top, it says 'Session Feedback' with a subtitle 'Share your experience for completed session record #1.' Below this is a white box titled 'Your Feedback'. It contains a 'Rating *' dropdown menu set to '5 - Very Satisfied', a 'Comments *' text area with the text 'The counsellor was supportive and the study plan was practical.', and two buttons at the bottom: 'Submit Feedback' (green) and 'Cancel' (blue).

Figure 3.4.22: Student Feedback Form

Figure 3.4.22 illustrates the Feedback Form for students to write after a counselling session is marked as "Completed." Students can rate their experience using a star-rating system and write comments in the provided text area. Clicking "Submit Feedback" saves these details into the session_records table.

3.4.23 Student Profile

The screenshot shows the 'My Profile' page for a student. On the left is a dark green sidebar with the 'Student Care' header and a list of navigation items: 'My Dashboard', 'Book Session', 'My Bookings', 'Session History', 'My Profile' (highlighted), 'Change Password', and 'Logout'. The main content area has a light green background. At the top, it says 'My Profile' and 'Update your personal account details.' Below this is a white rounded rectangle containing the 'Account Information' form. The form has two columns. The left column contains 'Username' (UK12345), 'Full Name *' (Nur Aina Rahman), and 'Phone Number *' (012-3456789). The right column contains 'Account Role' (Student) and 'Email *' (uk12345@gmail.com). At the bottom of the form are three buttons: 'Save Changes' (green), 'Change Password' (blue), and 'Cancel' (light blue).

Figure 3.4.23: Student My Profile Interface

Figure 3.4.23 presents the "My Profile" page for students. Similar to other roles, the "Username" and "Role" fields are read-only. Students can update their "Full Name", "Email" and "Phone Number." The interface includes options to save profile updates, navigate to the password change page or cancel the current operation.

3.4.24 Change Password (Student Portal)

The screenshot shows the 'Change Password' page for a student. On the left is a dark green sidebar with the 'Student Care' header and a list of navigation items: 'My Dashboard', 'Book Session', 'My Bookings', 'Session History', 'My Profile', 'Change Password' (highlighted), and 'Sign Out'. The main content area has a light green background. At the top, it says 'Change Password' and 'Use at least 6 characters, including one letter and one number. Spaces are not allowed.' Below this is a white rounded rectangle containing the 'Password Information' form. The form has three input fields: 'Current Password *', 'New Password *', and 'Confirm New Password *'. At the bottom of the form are two buttons: 'Update Password' (green) and 'Cancel' (light blue).

Figure 3.4.24: Student Change Password Interface

Figure 3.4.24 displays the "Change Password" interface for students. The form requires the user to input their "Current Password", "New Password", and "Confirm New Password." Same as

other role's portal, it also provides clear instructions on password requirements (minimum 6 characters, at least one letter and one number, no spaces).

3.4.25 Edit Session Record Page

The screenshot shows the 'Edit Session Record' page in the 'Counsellor Portal'. The left sidebar contains navigation links: Dashboard, Manage Availability, Manage Bookings, Session Records (highlighted), My Profile, and Logout. The main content area is titled 'Edit Session Record' with a subtitle 'Update the counselling session notes.' It features two sections: 'Session Information' and 'Counsellor Notes'. The 'Session Information' section contains four input fields: Student (Nur Aina Rahman), Student Email (uk12345@gmail.com), Session Date (2026-06-22), and Session Time (10:00:00). The 'Counsellor Notes' section has a text area with the content: 'The student discussed academic stress and difficulty managing study time. A weekly study plan and follow-up actions were agreed.' Below the text area is a note: 'These notes are for counsellor records. Students will not see them.' At the bottom of the notes section are two buttons: 'Update Notes' and 'Cancel'.

Figure 3.4.25: Edit Session Record Interface

Figure 3.4.25 displays the Edit Session Record interface which allows counsellors to modify previously recorded session documentation. The page displays non-editable "Session Information", which includes the student's name, email, the date of the session and the session time. The counsellor can update the "Session Notes" in the provided text area which will be kept confidential and are not visible to students. The "Update Notes" button saves the modifications to the database and the "Cancel" button allows the counsellor to return to the session records list without applying any changes.

4.0 Results

4.1 Results Obtained

Our system was tested using dummy Student, Counsellor and Administrator accounts stored in the MySQL database. The main counselling workflow can be completed from login until session record and feedback submission.

Test Scenario	Expected Result	Result
Login using Student ID	Student is redirected to Student Dashboard.	Pass
Login using Counsellor ID	Counsellor is redirected to Counsellor Dashboard.	Pass
Login using Admin ID	Administrator is redirected to Admin Dashboard.	Pass
View Student List	Only student records are displayed.	Pass
Edit Student phone number	Only phone number can be updated by administrator.	Pass
Add Counsellor	Counsellor profile is saved and Counsellor ID is generated from email.	Pass
Edit Counsellor	Counsellor details are updated successfully.	Pass
Delete Counsellor	Counsellor record is deleted successfully.	Pass
Create Availability Slot	Counsellor slot is saved successfully.	Pass
Create Overlapping Slot	System prevents conflicting slot.	Pass

Book Counselling Session	Student booking is created with pending status.	Pass
Reschedule Pending Booking	Students can select another available slot.	Pass
Approve or Reject Booking	Counsellor can update booking status.	Pass
Cancel Booking	Booking status changes to cancelled.	Pass
Create Session Record	Counsellor notes are stored successfully.	Pass
Submit Feedback	Student rating and feedback are stored successfully.	Pass
Invalid Phone Number	System displays validation errors.	Pass
Invalid Password	System displays validation errors.	Pass

Table 4.1.1 Result Obtained Table

Table 4.1.1 shows the result obtained from our system in difference scenario. We have 18 test scenarios including login, view student, manage availability, booking, session record and feedback and all are tested passed.

4.2 Discussion on whether project objectives have been achieved

Based on the system development and testing results, the objectives of this project have been achieved. The Counselling Session Booking System was successfully developed as a web-based system that can be used by students, counsellors and administrators. Each user role has its own dashboard and functions and they only can access features related to their responsibilities.

For students, the system provides a platform to view available counsellor schedules and book counselling sessions online. Students can also view their booking status, reschedule pending bookings, cancel bookings when necessary, view session history and submit feedback after completing a session. This reduces the need for students to contact counselors manually to ask about available dates and times.

For counselors, the system allows them to manage their availability schedules by creating, viewing, updating and deleting available slots. Counsellors can review booking requests and approve, reject or cancel bookings. They can also record counselling session notes after a session has been completed. This helps counsellors manage appointments and session information in a more organised way.

The administrator functions also achieved the intended objectives. Administrators can manage counsellor information through add, view, update and delete functions. They can also monitor booking records. Student accounts are prepared as dummy data in the database (since we are developing for the UMT counseling system) while administrators can view the Student List and update student phone numbers when needed.

Our system also provides centralized storage for user information, schedules, bookings, session records, ratings and feedback through the MySQL database. This really makes the information easier to retrieve for reference and monitoring purposes. As we can see, compared to manual methods such as messaging, calls or separate records, our system improves productivity, reduces repeated work and improves data accuracy.

Finally, validation functions were implemented too to reduce invalid input and schedule conflicts. The system validates required fields, email format, Malaysian phone numbers, passwords, ratings, feedback, future dates and schedule overlap. Therefore, our system has achieved the main project objectives that have been planned.

5.0 Conclusion

5.1 Summary of the project

In summary, the Counseling Session Booking System project has been successfully developed as a web-based application that can assist in the management of counseling sessions in a more systematic, efficient and organized manner. This system was developed to overcome various problems that often occur in the manual booking process such as overlapping schedules, difficulty in managing records, loss of important information and delays in the communication process between students and counselors.

Through the development of this system, students can now book counseling sessions online without having to contact counselors directly to get available slots. In addition, counselors can manage their availability schedules more easily and record counseling session information regularly. Administrators can also manage users and counselors through complete data management functions.

This project was developed using Java Web, JSP, Servlet, HTML, CSS, JavaScript and MySQL technologies by applying Model-View-Controller (MVC) architecture. The use of this technology helps produce a more organized system, easy to maintain and able to support future system expansion.

All objectives that were set at the initial stage of the project were successfully achieved. This system not only successfully provided a well-functioning counseling session booking platform but also improved data management efficiency, reduced human error and provided a better user experience for all parties involved.

In addition to fulfilling the academic requirements for the CSE3023 Web-Based Application Development course, this project also provided valuable experience to the group members in web application development, database design, user interface integration, CRUD function development and system testing. This experience helped improve technical skills and teamwork which are very important in real software development.

5.2 Strengths and limitations

System Strengths :

1. Provides a centralized platform for counseling session management.
2. Easy-to-use and user-friendly interface.
3. Reduces errors in booking and schedule management.
4. Systematically stores user data, bookings and session records.
5. Uses MVC architecture that simplifies system maintenance.

System Limitations

1. Can only be accessed via web browser.
2. No automatic notification function such as email or SMS.
3. Does not yet support online counseling sessions.
4. Reporting and data analysis functions are still limited

5.3 Future improvements

Future improvements for the effectiveness of the system in the future, several improvements can be implemented. Among them is the development of a mobile application so that users can access the system more easily via smartphones. In addition, automatic notification functions via email or SMS can be added to remind students and counselors about upcoming sessions, thus helping to reduce absenteeism rates.

In addition, the system can be enhanced by integrating video conferencing platforms such as Google Meet or Zoom to enable counseling sessions to be conducted online. Reporting and data analysis functions can also be added to help administrators monitor the performance of counseling services more effectively. The addition of security features such as two-factor authentication (2FA) can also be considered to ensure that user data and accounts are more secure.

6.0 References

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